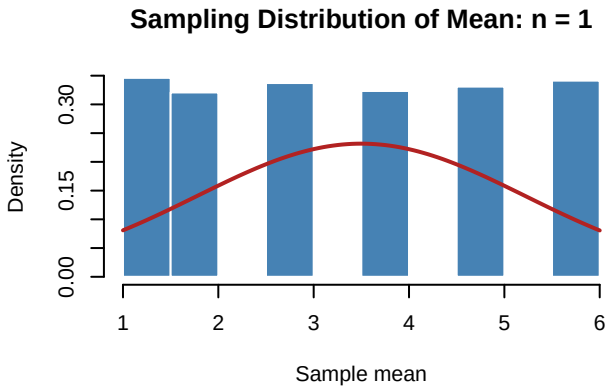
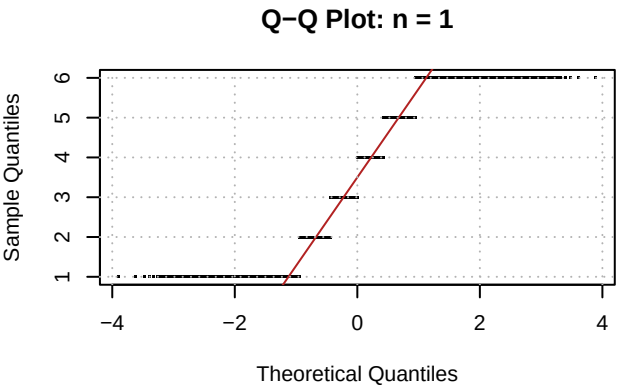
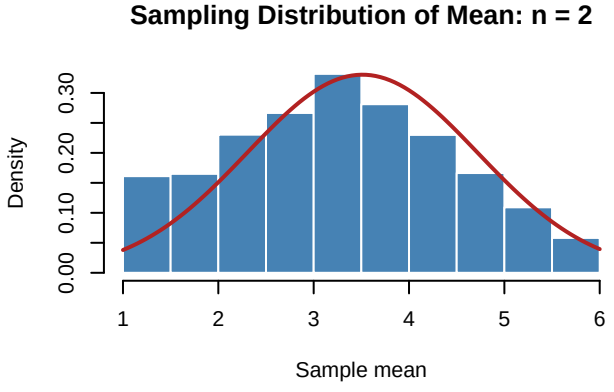
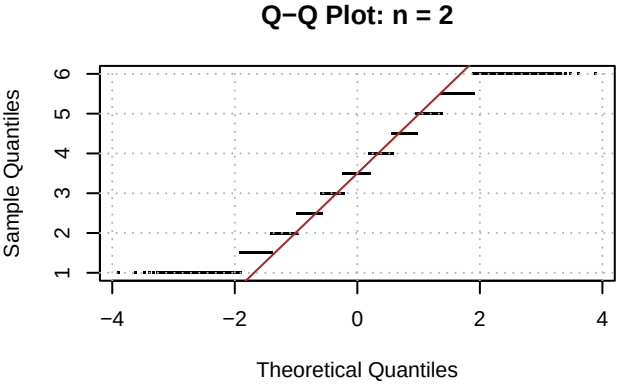
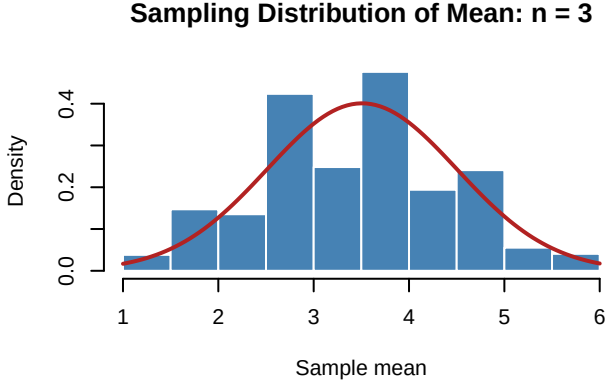
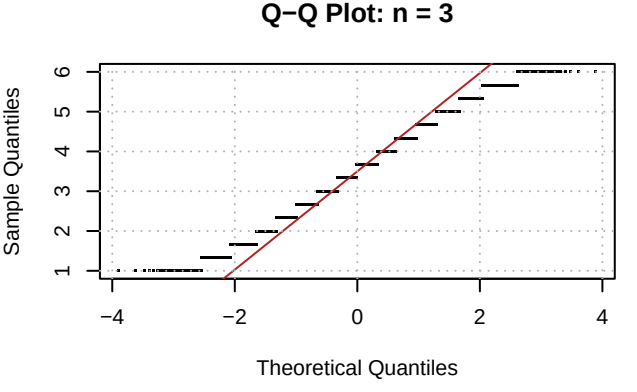

Sampling Distribution of Mean: n = 1
Density
0.00
0.15
0.30
1
2
3
4
5
6
Sample mean

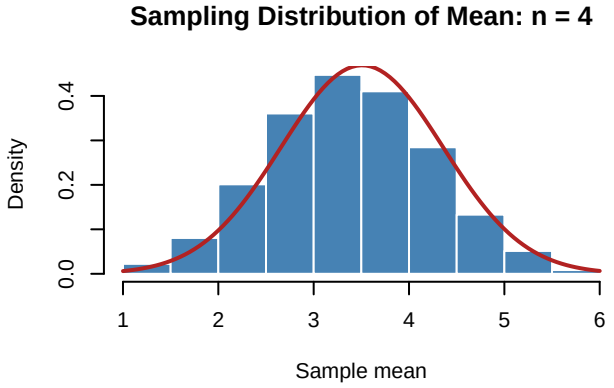

Q−Q Plot: n = 1
Sample Quantiles
1
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

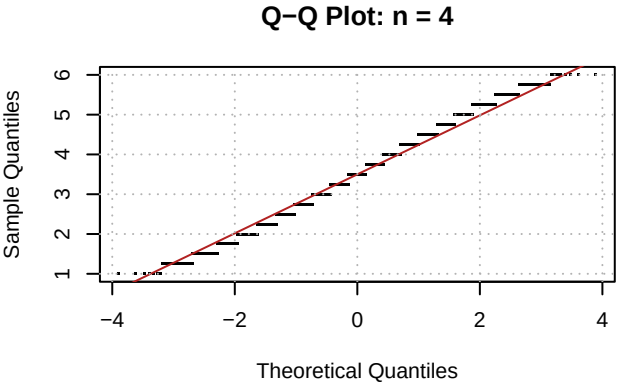

Sampling Distribution of Mean: n = 2
Density
0.00
0.10
0.20
0.30
1
2
3
4
5
6
Sample mean

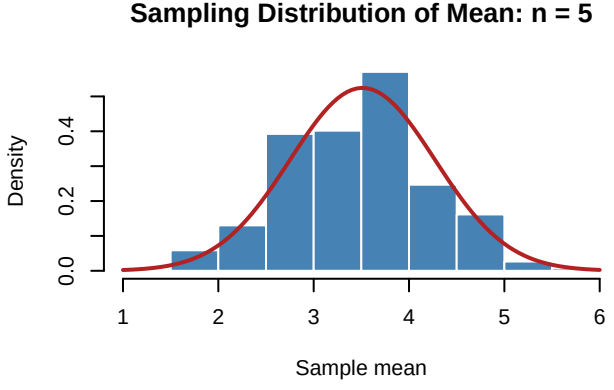

Q−Q Plot: n = 2
Sample Quantiles
1
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

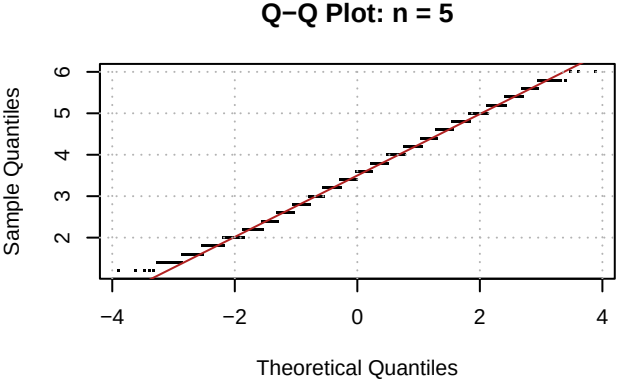

Sampling Distribution of Mean: n = 3
Density
0.0
0.2
0.4
1
2
3
4
5
6
Sample mean

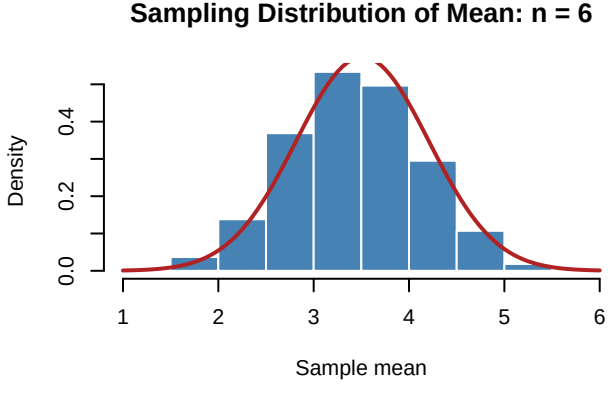

Q−Q Plot: n = 3
Sample Quantiles
1
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

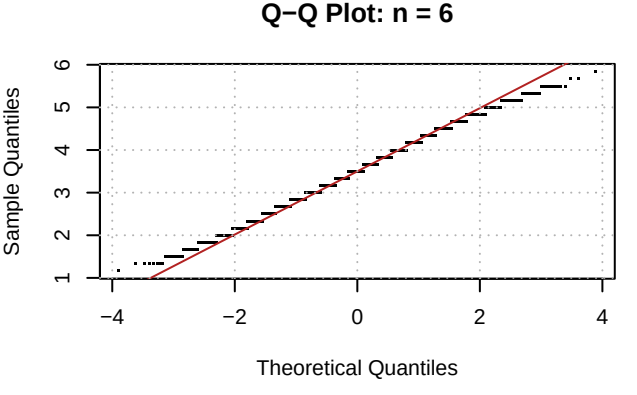

Sampling Distribution of Mean: n = 4
Density
0.0
0.2
0.4
1
2
3
4
5
6
Sample mean

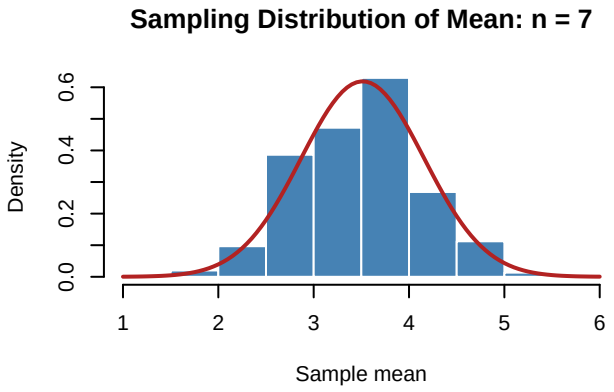

Q−Q Plot: n = 4
Sample Quantiles
1
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

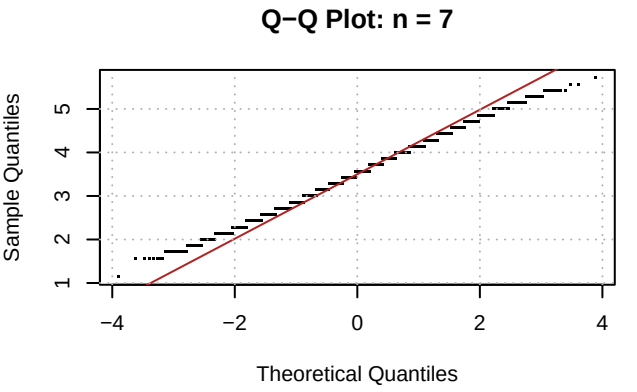

Sampling Distribution of Mean: n = 5
Density
0.0
0.2
0.4
1
2
3
4
5
6
Sample mean

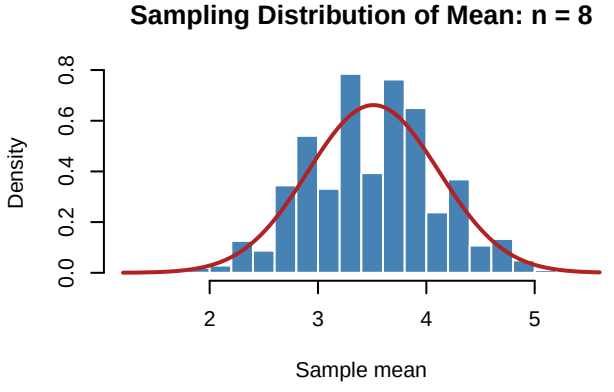

Q−Q Plot: n = 5
Sample Quantiles
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

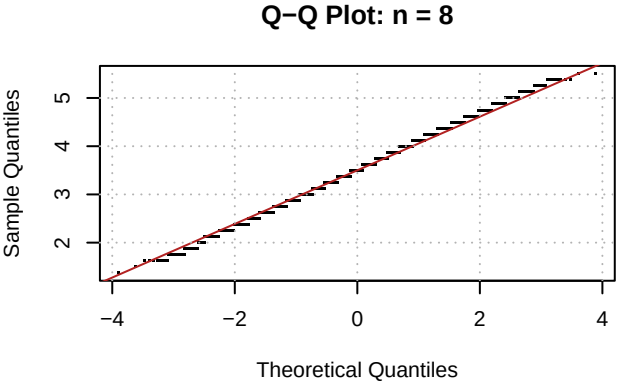

Sampling Distribution of Mean: n = 6
Density
0.0
0.2
0.4
1
2
3
4
5
6
Sample mean

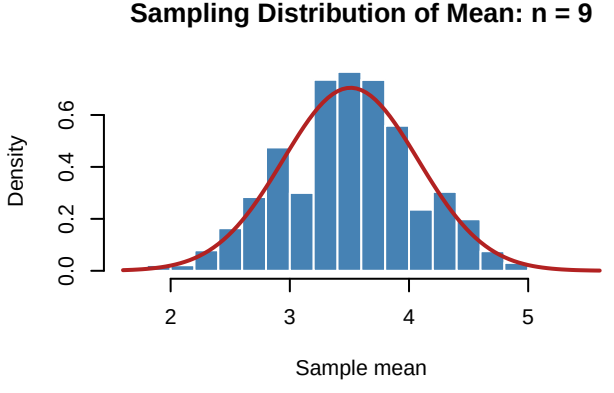

Q−Q Plot: n = 6
Sample Quantiles
1
2
3
4
5
6
−4
−2
0
2
4
Theoretical Quantiles

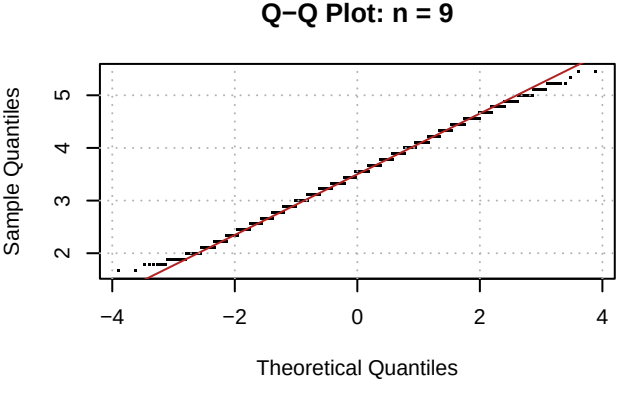

Sampling Distribution of Mean: n = 7
Density
0.0
0.2
0.4
0.6
1
2
3
4
5
6
Sample mean

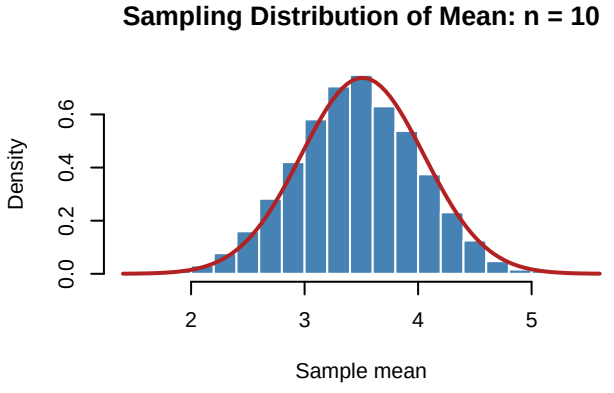

Q−Q Plot: n = 7
Sample Quantiles
1
2
3
4
5
−4
−2
0
2
4
Theoretical Quantiles

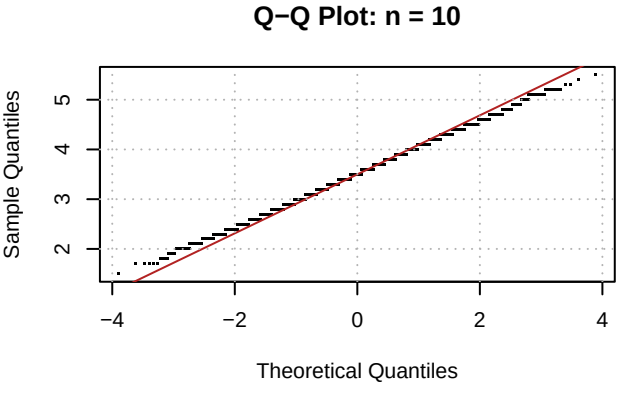

Sampling Distribution of Mean: n = 8
Density
0.0
0.2
0.4
0.6
0.8
2
3
4
5
Sample mean

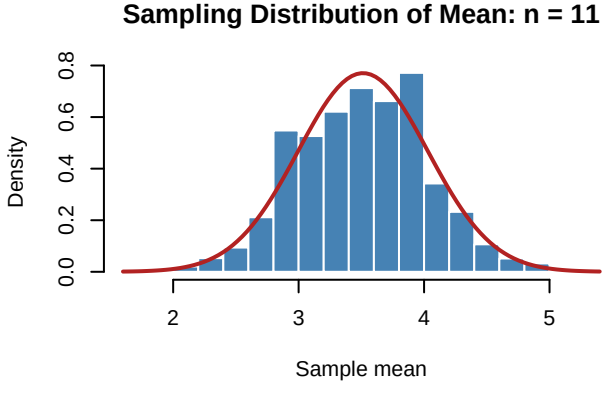

Q−Q Plot: n = 8
Sample Quantiles
2
3
4
5
−4
−2
0
2
4
Theoretical Quantiles

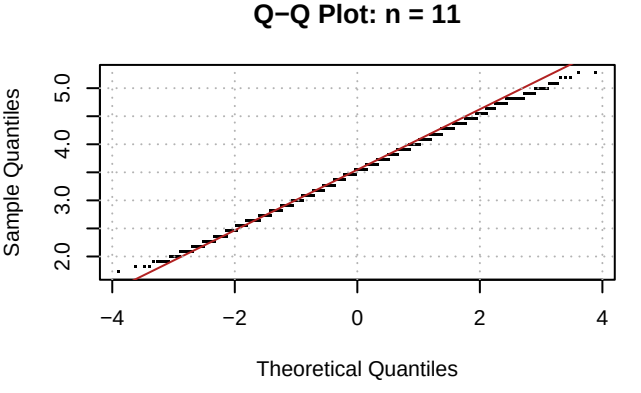

Sampling Distribution of Mean: n = 9
Density
0.0
0.2
0.4
0.6
2
3
4
5
Sample mean

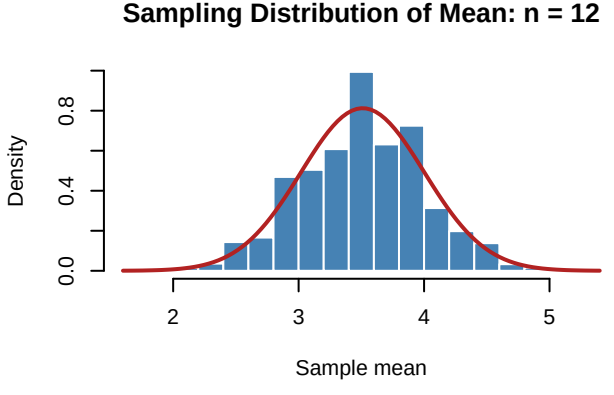

Q−Q Plot: n = 9
Sample Quantiles
2
3
4
5
−4
−2
0
2
4
Theoretical Quantiles

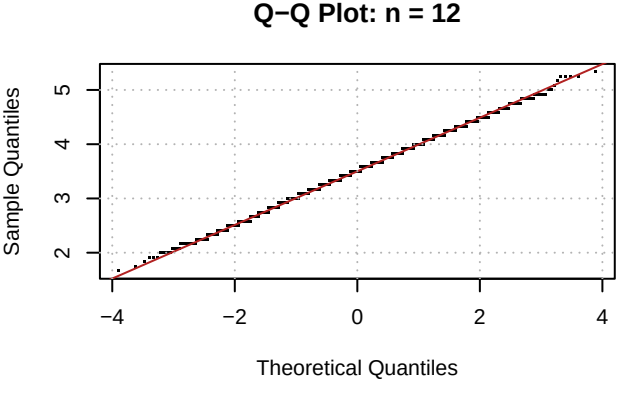

Sampling Distribution of Mean: n = 10
Density
0.0
0.2
0.4
0.6
2
3
4
5
Sample mean

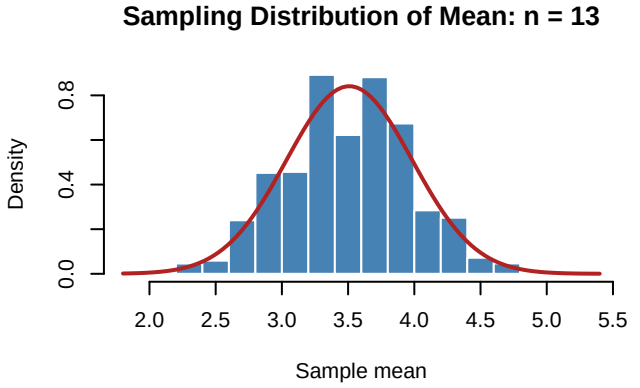

Q−Q Plot: n = 10
Sample Quantiles
2
3
4
5
−4
−2
0
2
4
Theoretical Quantiles

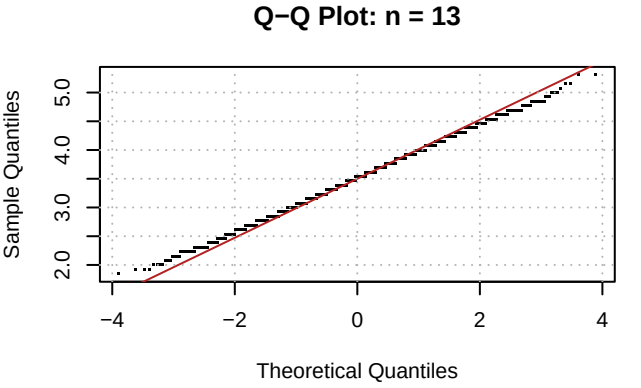

Sampling Distribution of Mean: n = 11
Density
0.0
0.2
0.4
0.6
0.8
2
3
4
5
Sample mean

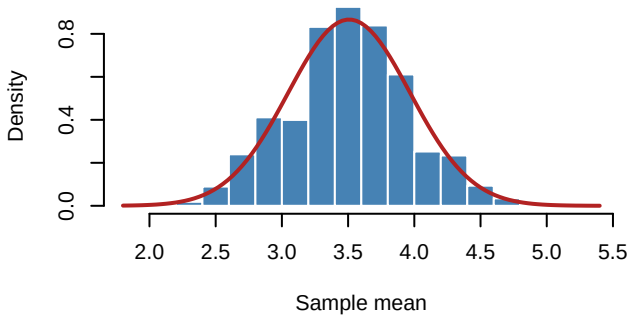

Q−Q Plot: n = 11
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

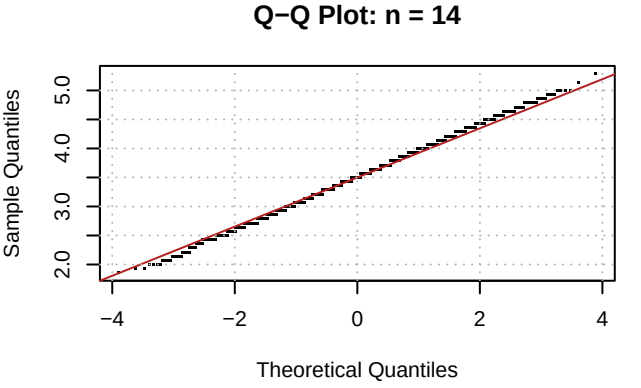

Sampling Distribution of Mean: n = 12
Density
0.0
0.4
0.8
2
3
4
5
Sample mean

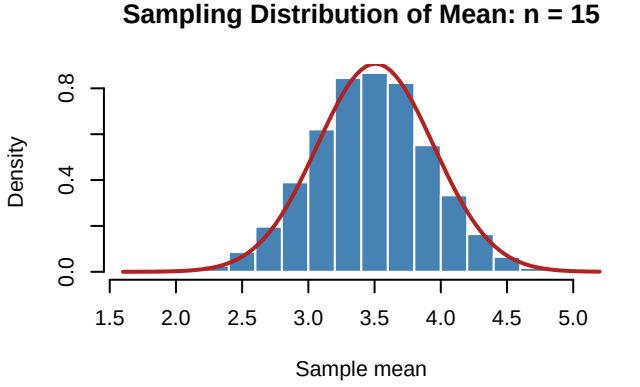

Q−Q Plot: n = 12
Sample Quantiles
2
3
4
5
−4
−2
0
2
4
Theoretical Quantiles

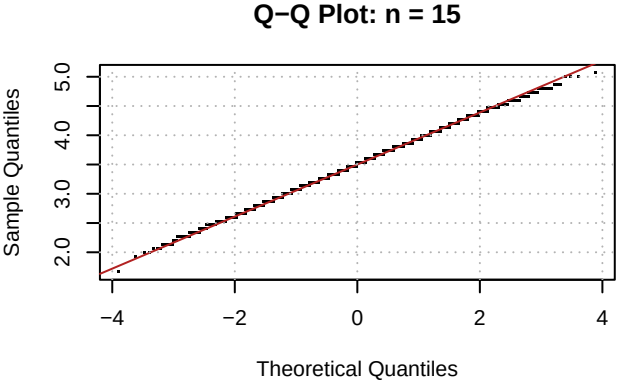

Sampling Distribution of Mean: n = 13
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
Sample mean

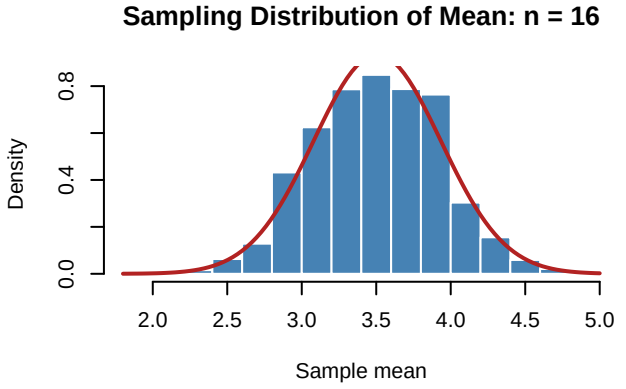

Q−Q Plot: n = 13
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

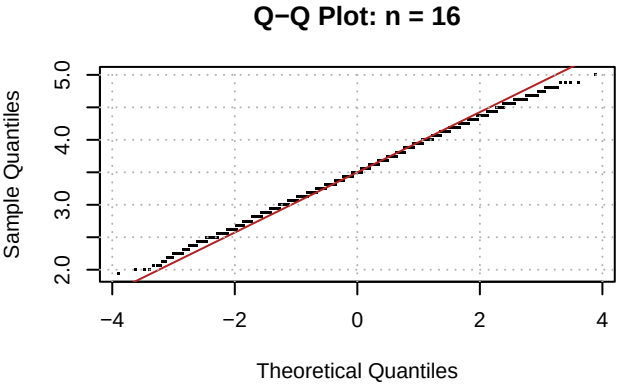

Sampling Distribution of Mean: n = 14
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
Sample mean

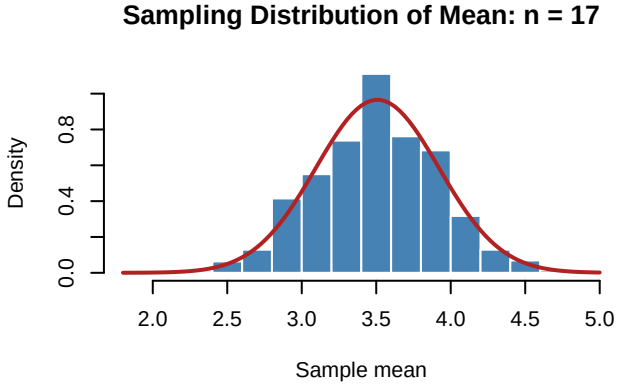

Q−Q Plot: n = 14
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

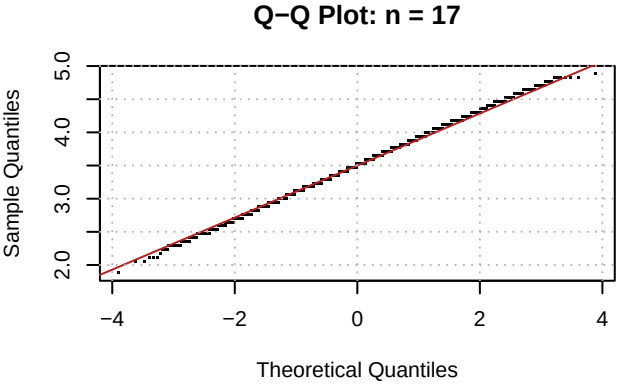

Sampling Distribution of Mean: n = 15
Density
0.0
0.4
0.8
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

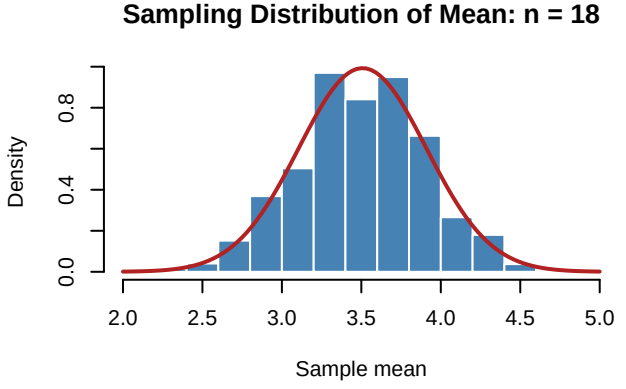

Q−Q Plot: n = 15
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

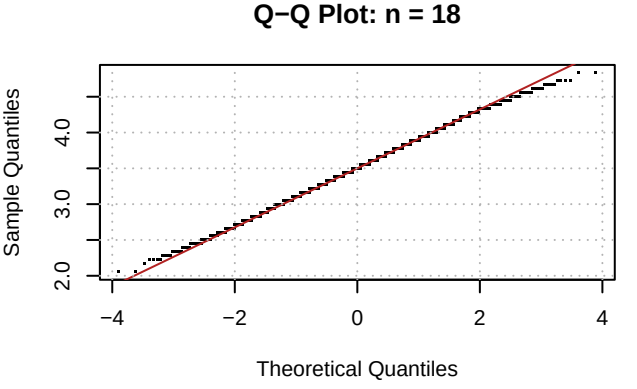

Sampling Distribution of Mean: n = 16
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

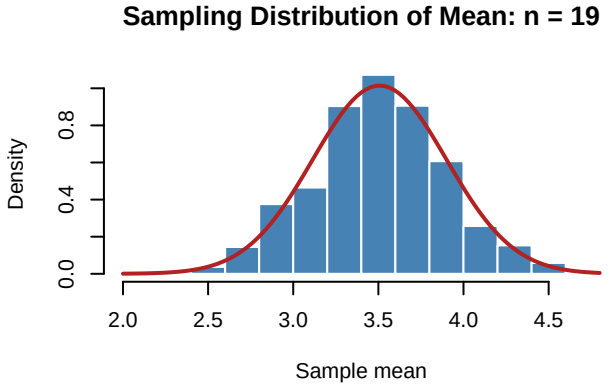

Q−Q Plot: n = 16
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

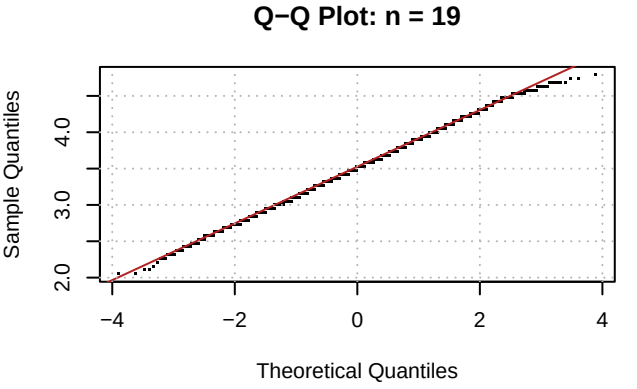

Sampling Distribution of Mean: n = 17
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

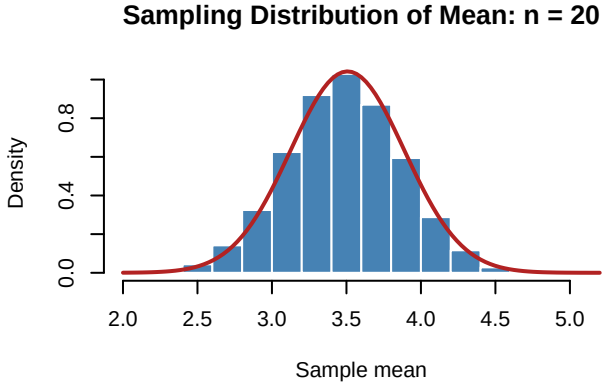

Q−Q Plot: n = 17
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

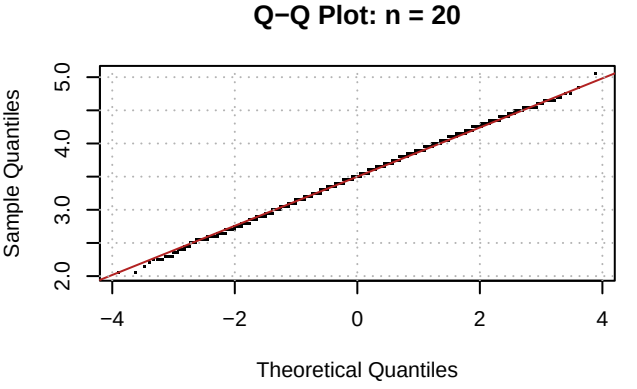

Sampling Distribution of Mean: n = 18
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

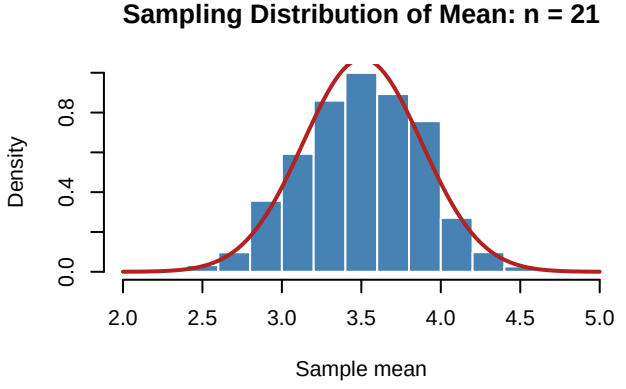

Q−Q Plot: n = 18
Sample Quantiles
2.0
3.0
4.0
−4
−2
0
2
4
Theoretical Quantiles

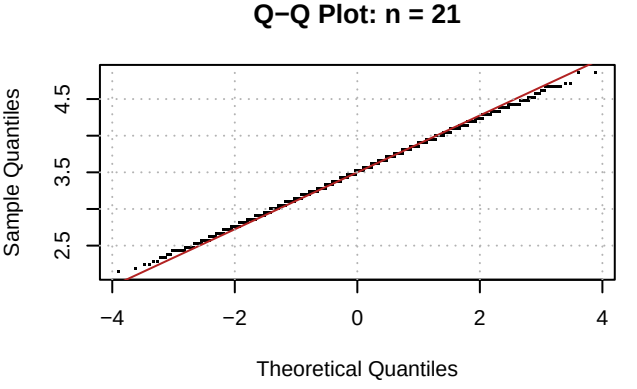

Sampling Distribution of Mean: n = 19
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
Sample mean

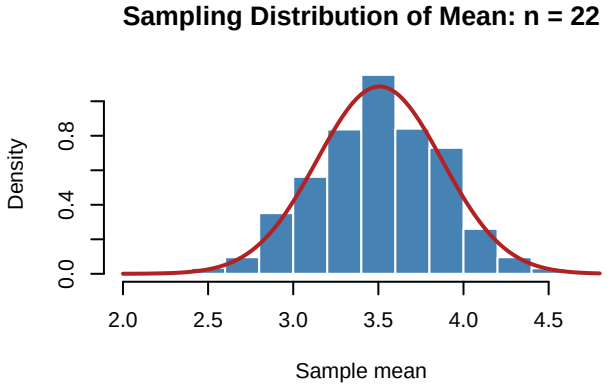

Q−Q Plot: n = 19
Sample Quantiles
2.0
3.0
4.0
−4
−2
0
2
4
Theoretical Quantiles

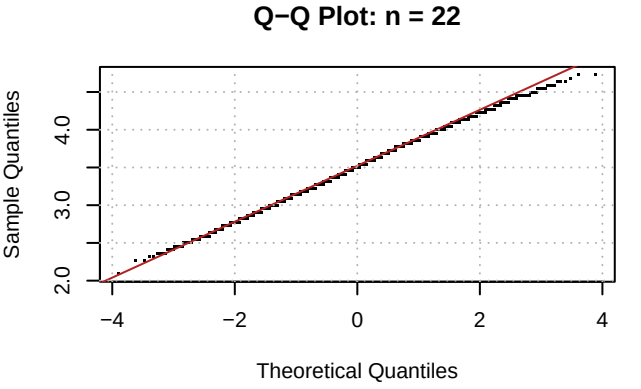

Sampling Distribution of Mean: n = 20
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

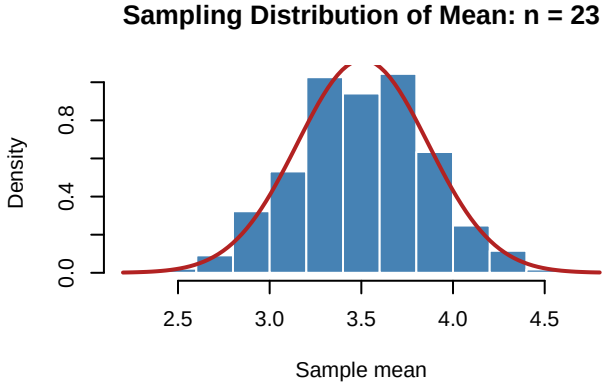

Q−Q Plot: n = 20
Sample Quantiles
2.0
3.0
4.0
5.0
−4
−2
0
2
4
Theoretical Quantiles

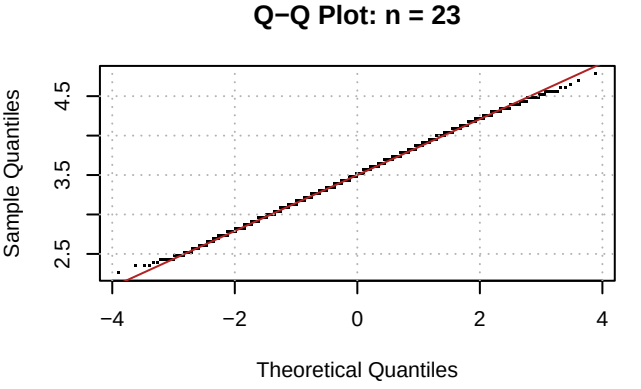

Sampling Distribution of Mean: n = 21
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

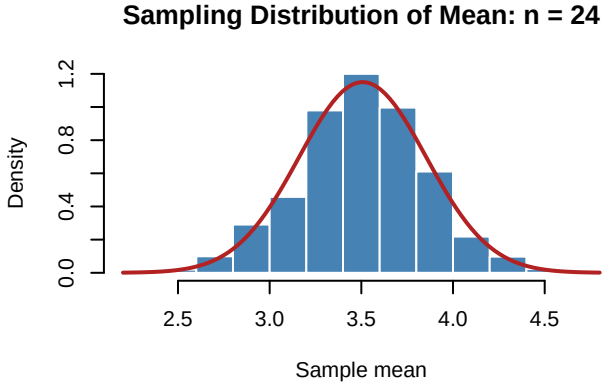

Q−Q Plot: n = 21
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

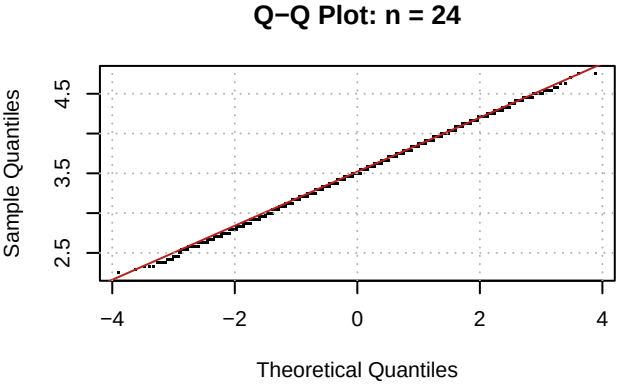

Sampling Distribution of Mean: n = 22
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
Sample mean

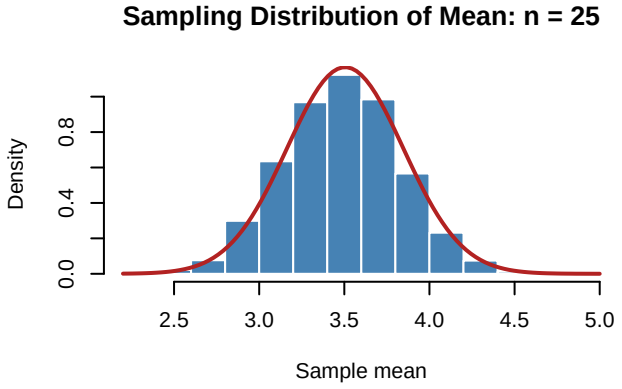

Q−Q Plot: n = 22
Sample Quantiles
2.0
3.0
4.0
−4
−2
0
2
4
Theoretical Quantiles

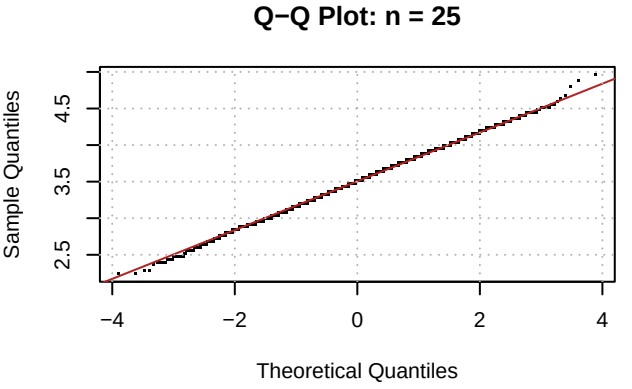

Sampling Distribution of Mean: n = 23
Density
0.0
0.4
0.8
2.5
3.0
3.5
4.0
4.5
Sample mean

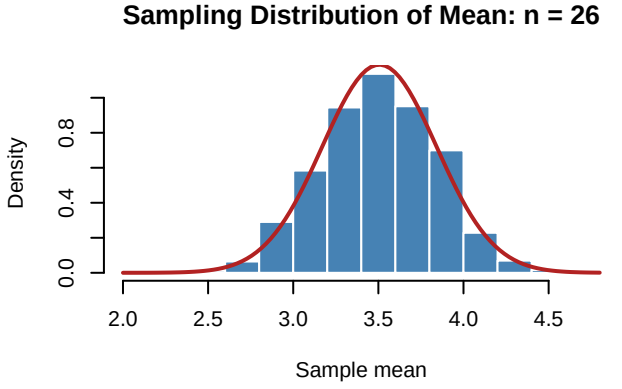

Q−Q Plot: n = 23
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

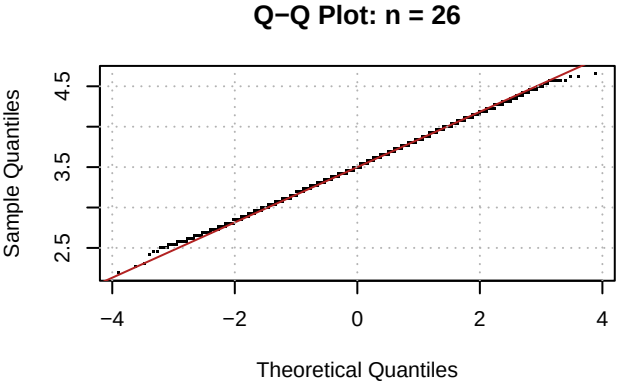

Sampling Distribution of Mean: n = 24
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

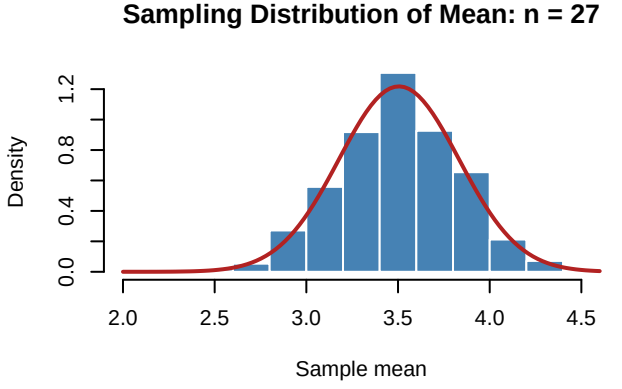

Q−Q Plot: n = 24
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

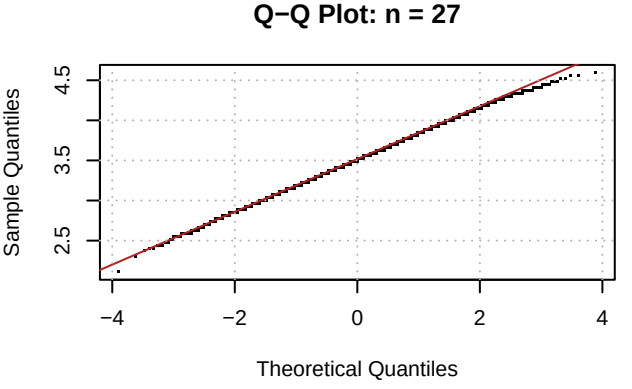

Sampling Distribution of Mean: n = 25
Density
0.0
0.4
0.8
2.5
3.0
3.5
4.0
4.5
5.0
Sample mean

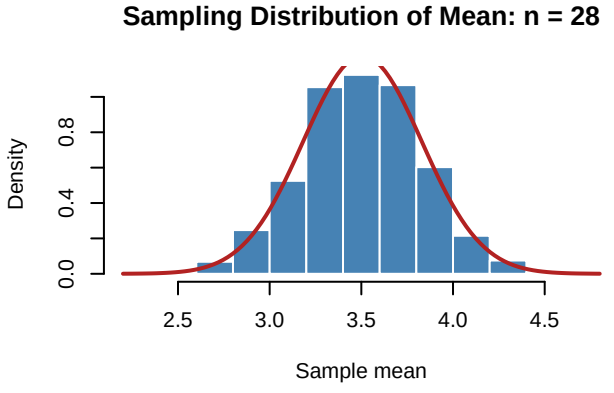

Q−Q Plot: n = 25
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

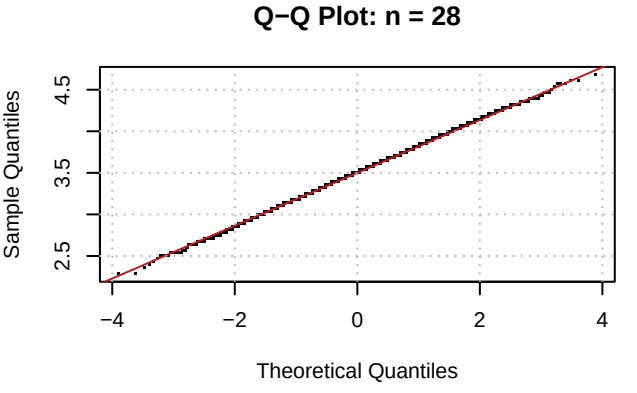

Sampling Distribution of Mean: n = 26
Density
0.0
0.4
0.8
2.0
2.5
3.0
3.5
4.0
4.5
Sample mean

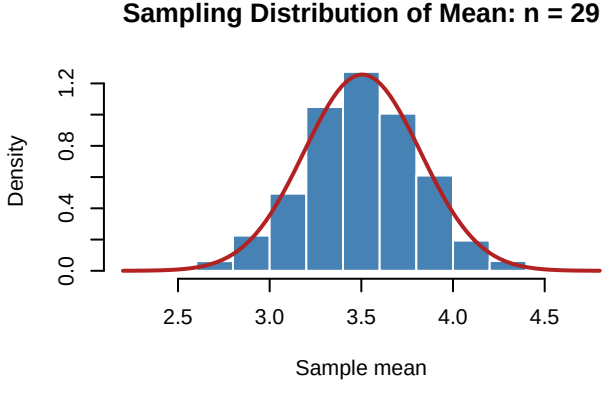

Q−Q Plot: n = 26
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

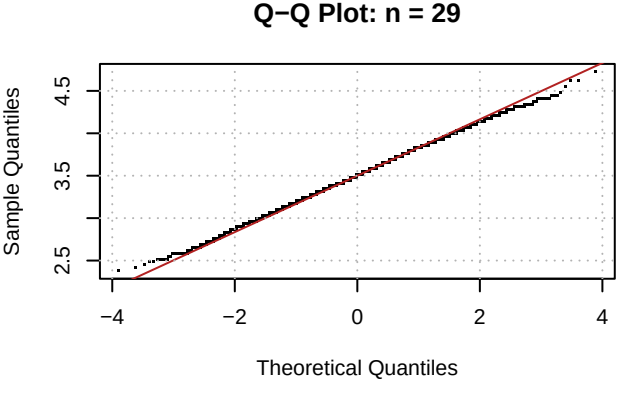

Sampling Distribution of Mean: n = 27
Density
0.0
0.4
0.8
1.2
2.0
2.5
3.0
3.5
4.0
4.5
Sample mean

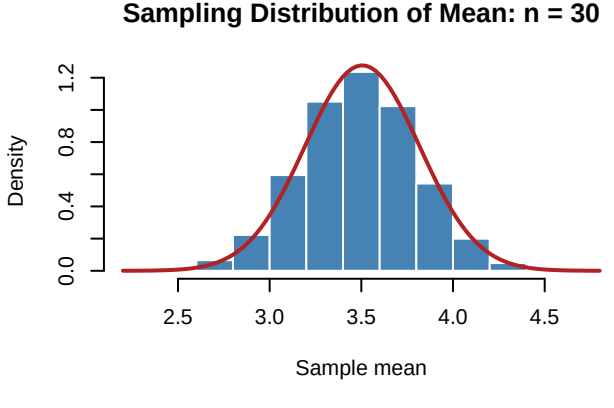

Q−Q Plot: n = 27
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

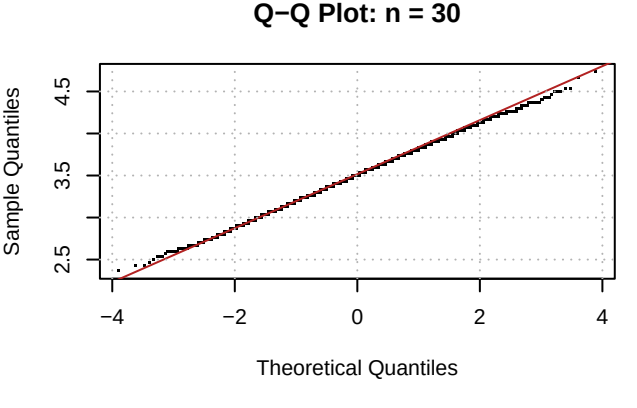

Sampling Distribution of Mean: n = 28
Density
0.0
0.4
0.8
2.5
3.0
3.5
4.0
4.5
Sample mean

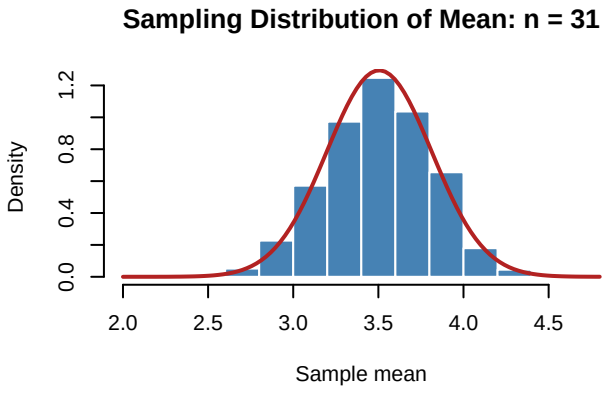

Q−Q Plot: n = 28
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

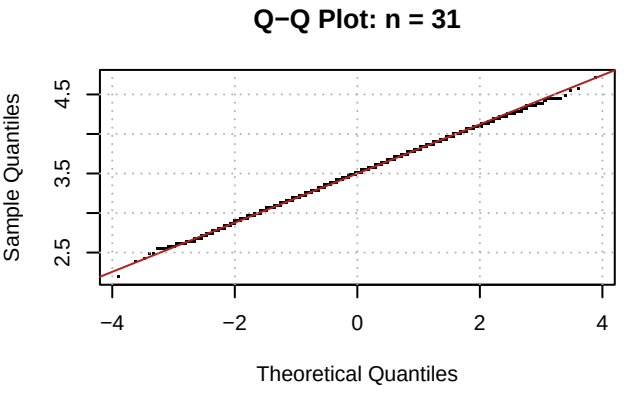

Sampling Distribution of Mean: n = 29
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

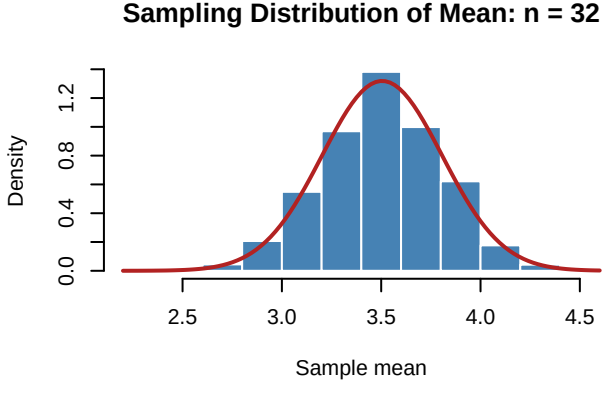

Q−Q Plot: n = 29
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

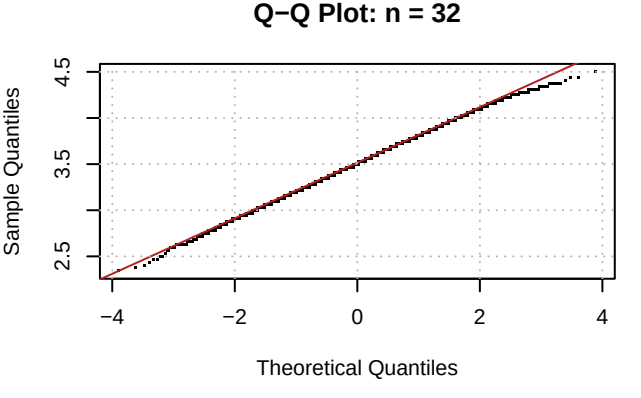

Sampling Distribution of Mean: n = 30
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

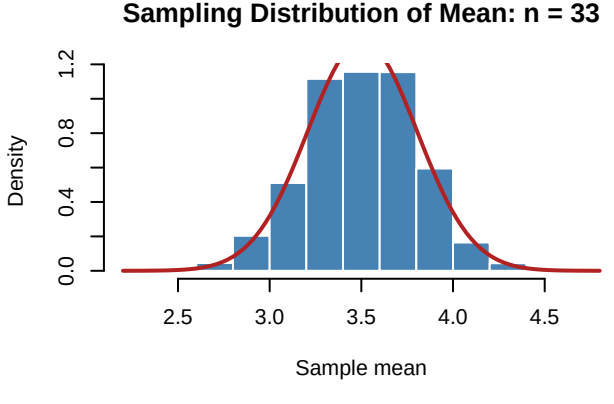

Q−Q Plot: n = 30
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

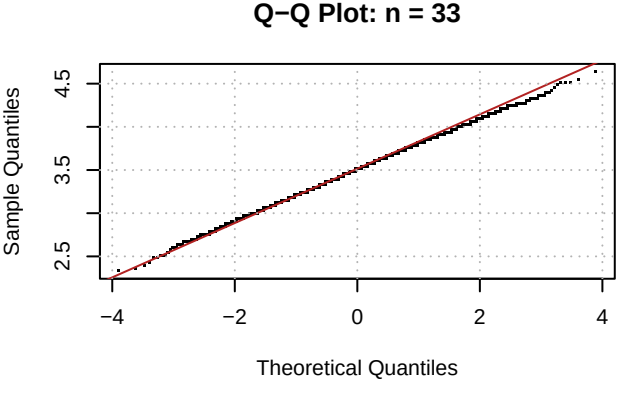

Sampling Distribution of Mean: n = 31
Density
0.0
0.4
0.8
1.2
2.0
2.5
3.0
3.5
4.0
4.5
Sample mean

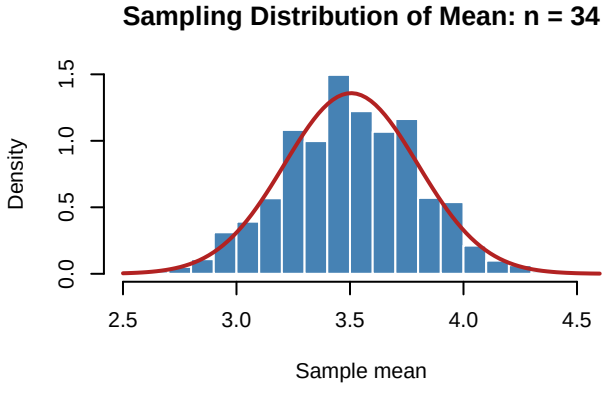

Q−Q Plot: n = 31
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

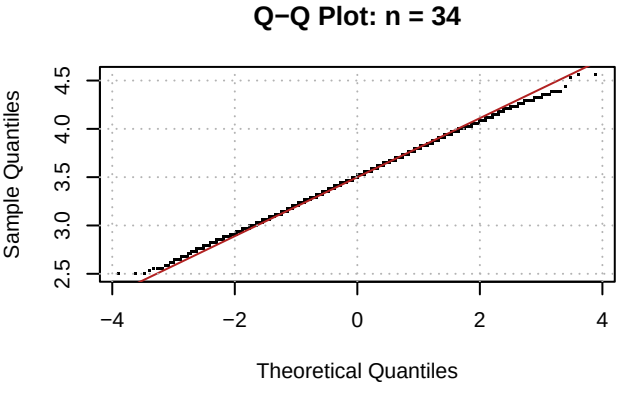

Sampling Distribution of Mean: n = 32
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

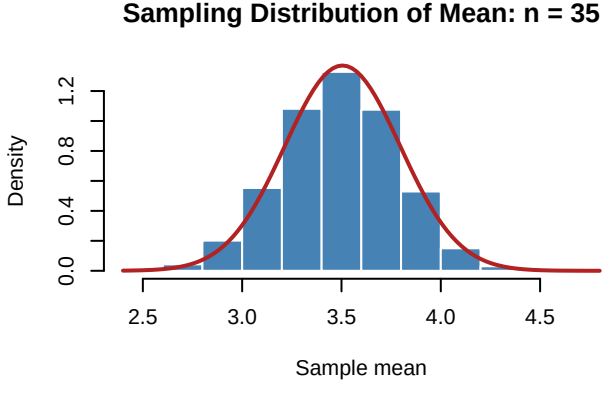

Q−Q Plot: n = 32
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

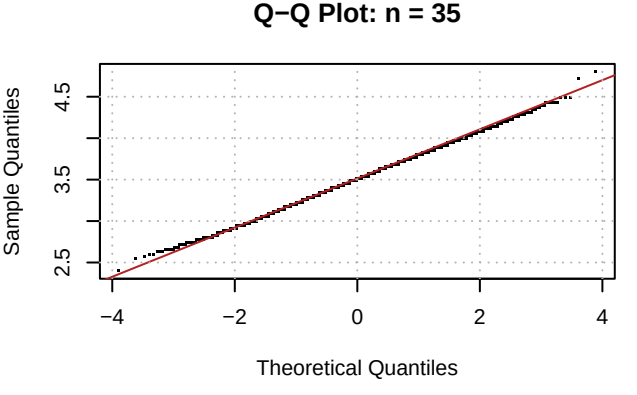

Sampling Distribution of Mean: n = 33
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

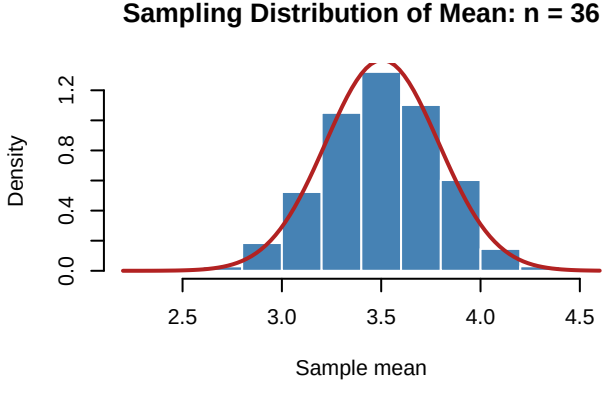

Q−Q Plot: n = 33
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

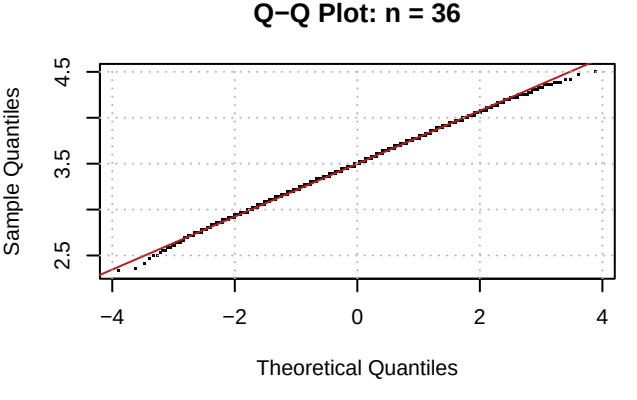

Sampling Distribution of Mean: n = 34
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

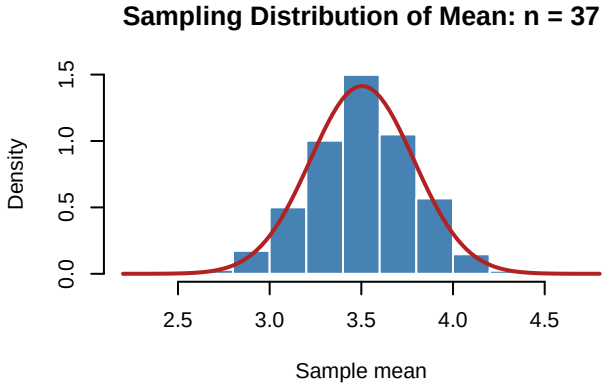

Q−Q Plot: n = 34
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

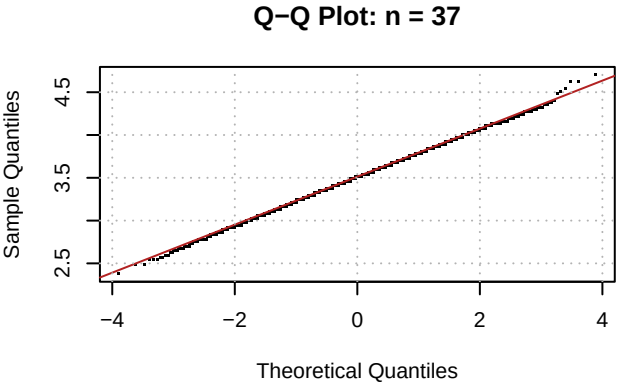

Sampling Distribution of Mean: n = 35
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

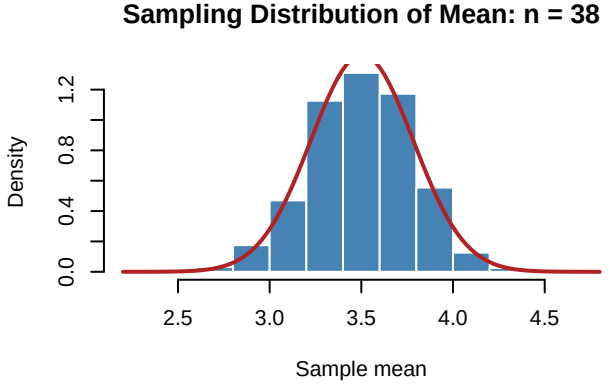

Q−Q Plot: n = 35
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

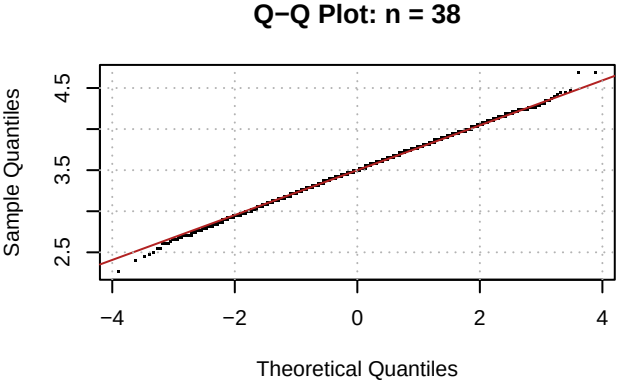

Sampling Distribution of Mean: n = 36
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

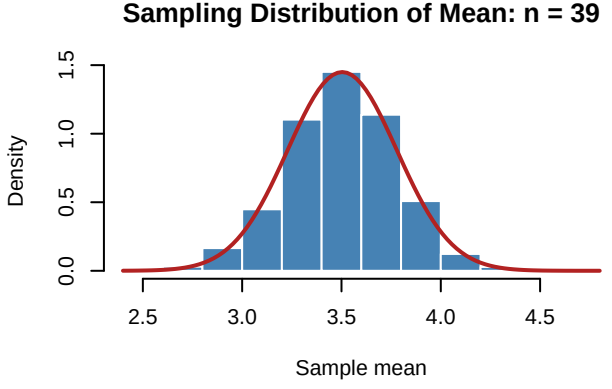

Q−Q Plot: n = 36
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

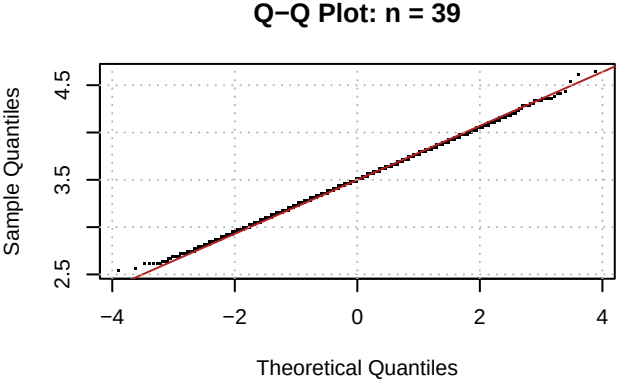

Sampling Distribution of Mean: n = 37
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

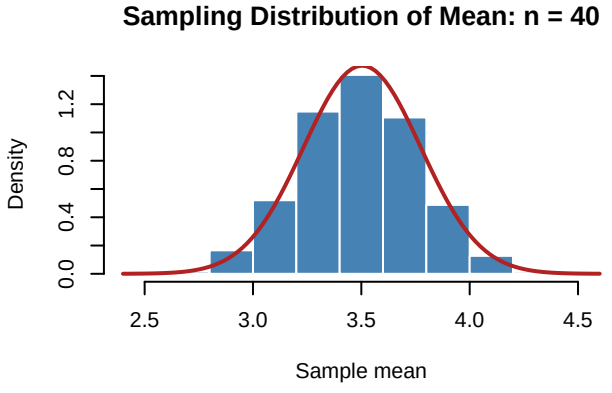

Q−Q Plot: n = 37
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

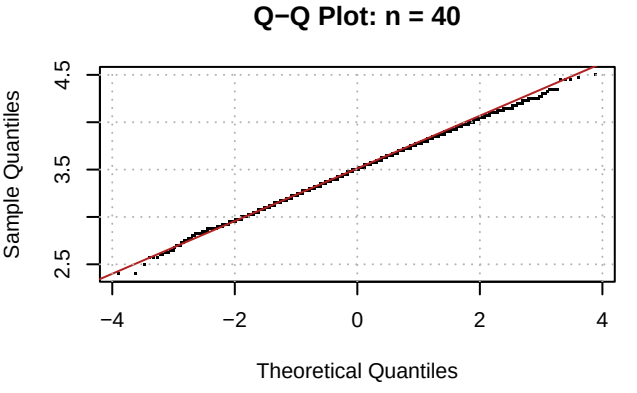

Sampling Distribution of Mean: n = 38
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

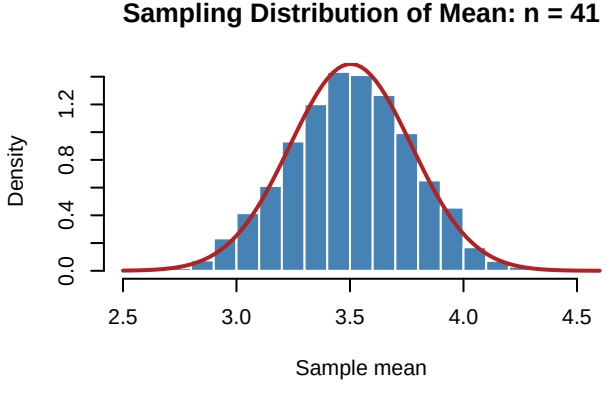

Q−Q Plot: n = 38
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

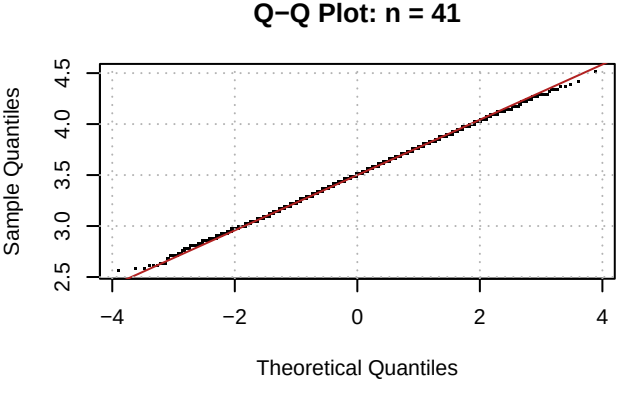

Sampling Distribution of Mean: n = 39
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

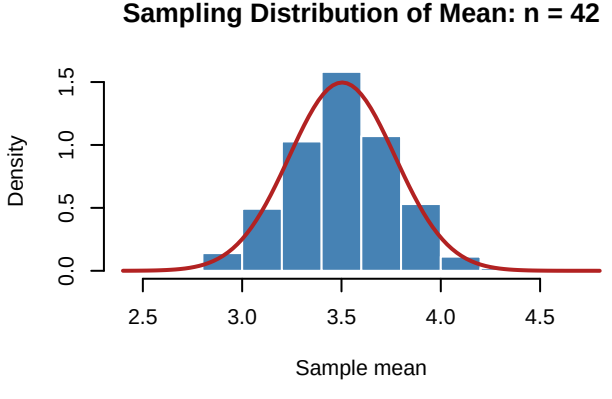

Q−Q Plot: n = 39
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

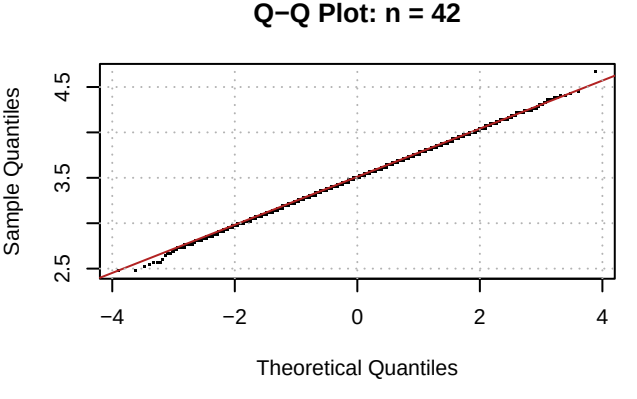

Sampling Distribution of Mean: n = 40
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

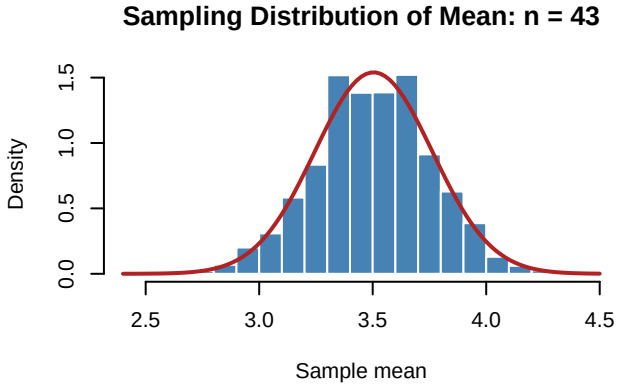

Q−Q Plot: n = 40
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

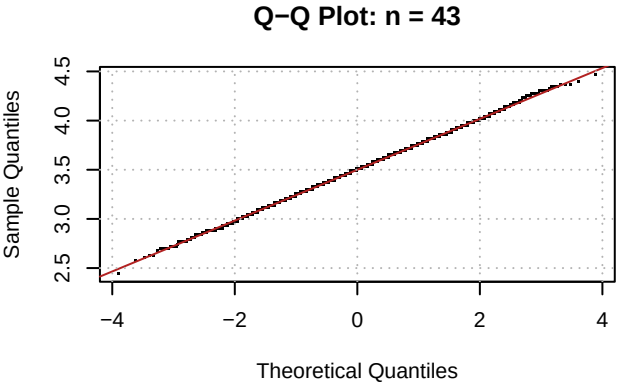

Sampling Distribution of Mean: n = 41
Density
0.0
0.4
0.8
1.2
2.5
3.0
3.5
4.0
4.5
Sample mean

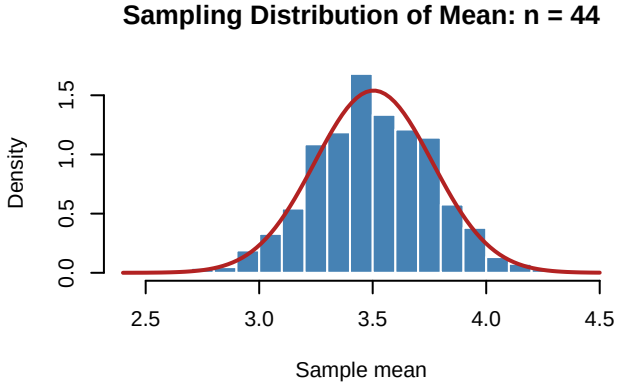

Q−Q Plot: n = 41
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

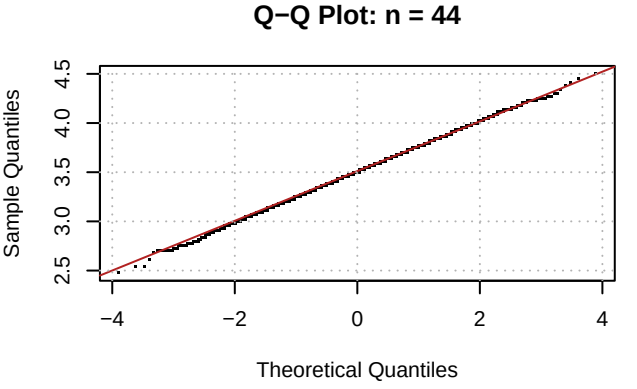

Sampling Distribution of Mean: n = 42
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

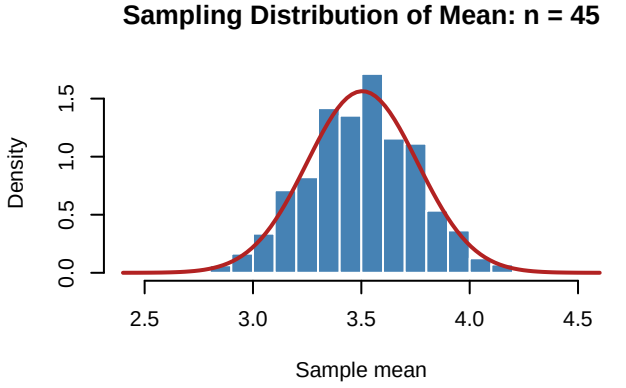

Q−Q Plot: n = 42
Sample Quantiles
2.5
3.5
4.5
−4
−2
0
2
4
Theoretical Quantiles

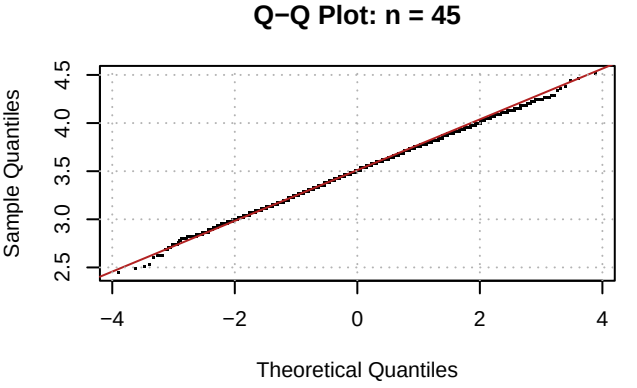

Sampling Distribution of Mean: n = 43
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

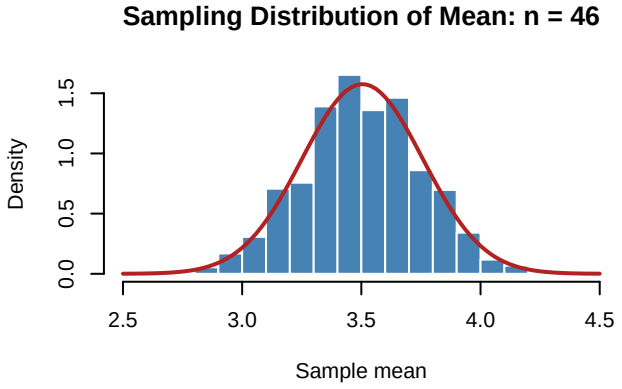

Q−Q Plot: n = 43
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

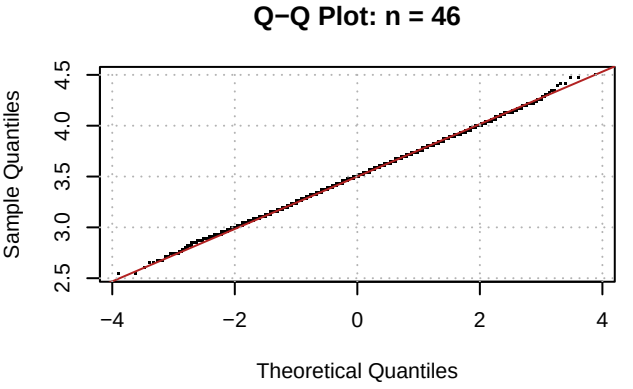

Sampling Distribution of Mean: n = 44
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

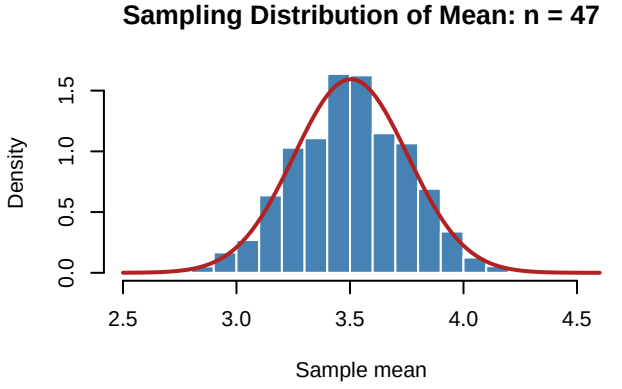

Q−Q Plot: n = 44
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

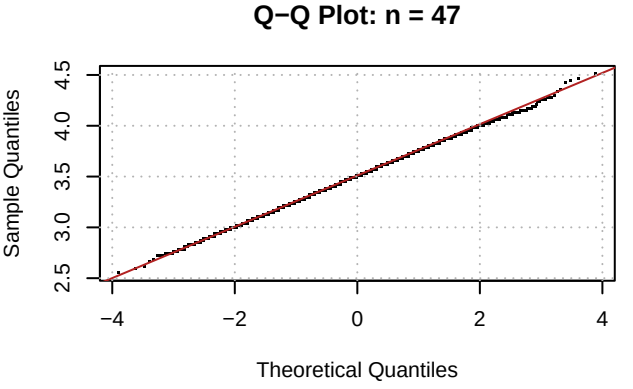

Sampling Distribution of Mean: n = 45
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

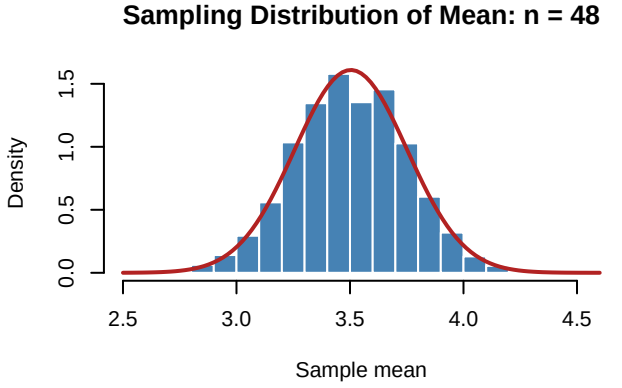

Q−Q Plot: n = 45
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

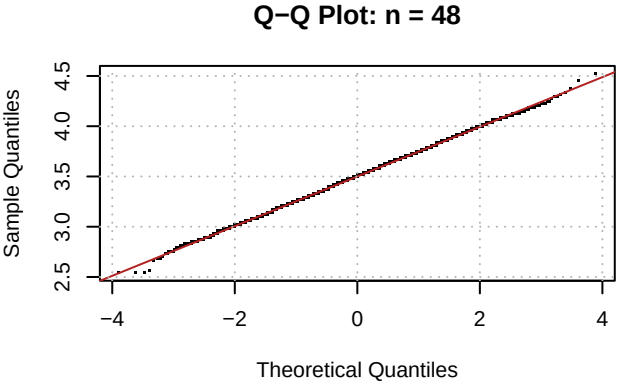

Sampling Distribution of Mean: n = 46
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

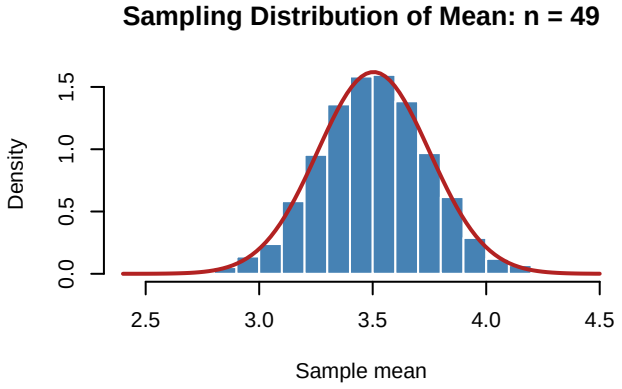

Q−Q Plot: n = 46
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

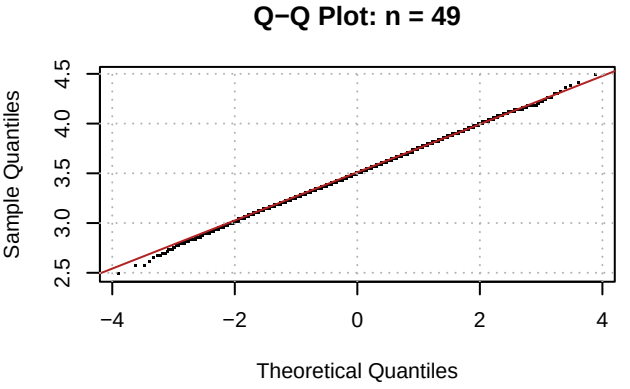

Sampling Distribution of Mean: n = 47
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

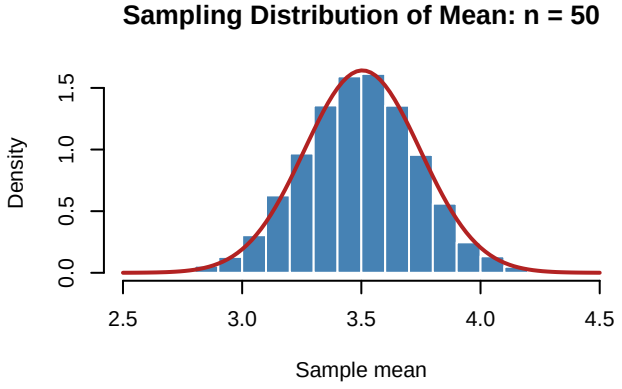

Q−Q Plot: n = 47
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

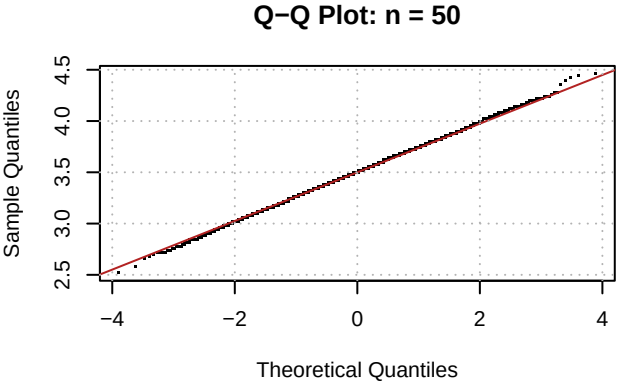

Sampling Distribution of Mean: n = 48
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

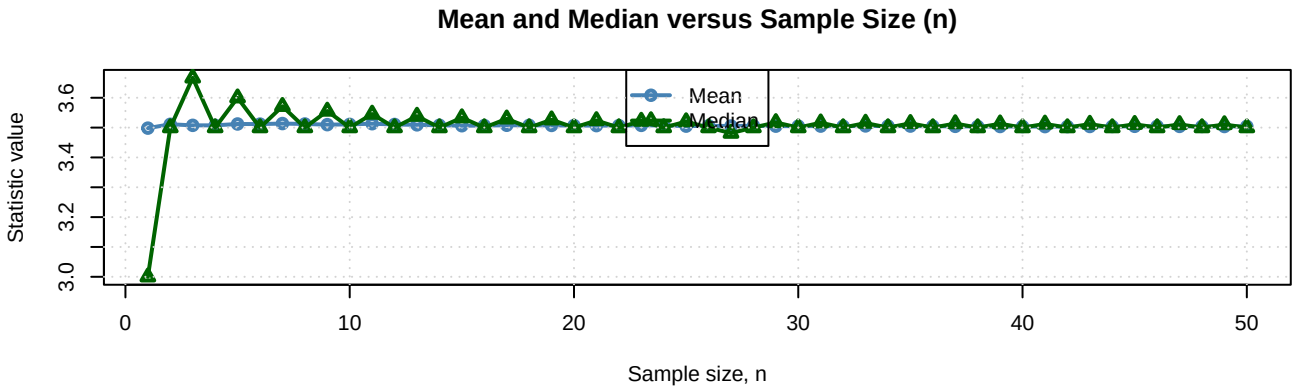

Q−Q Plot: n = 48
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles

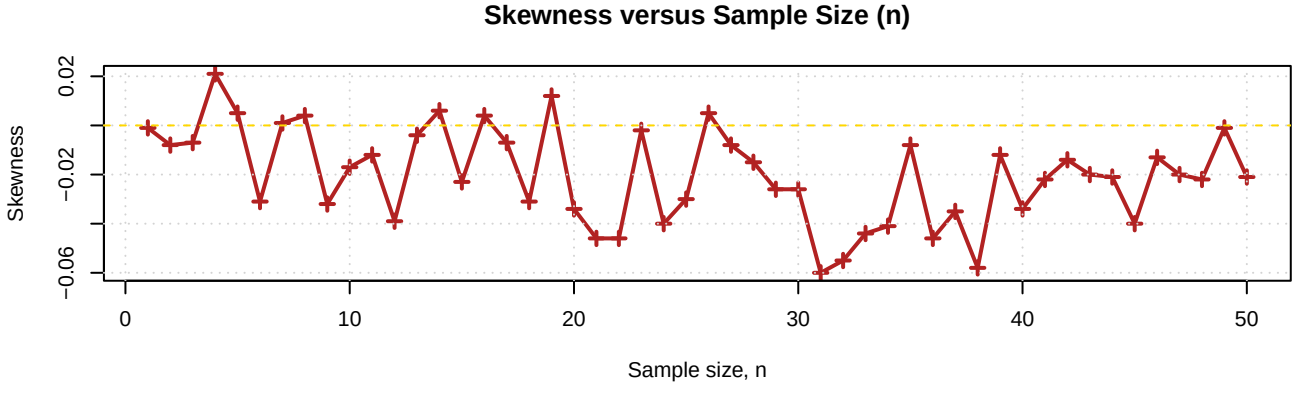

Sampling Distribution of Mean: n = 49
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean

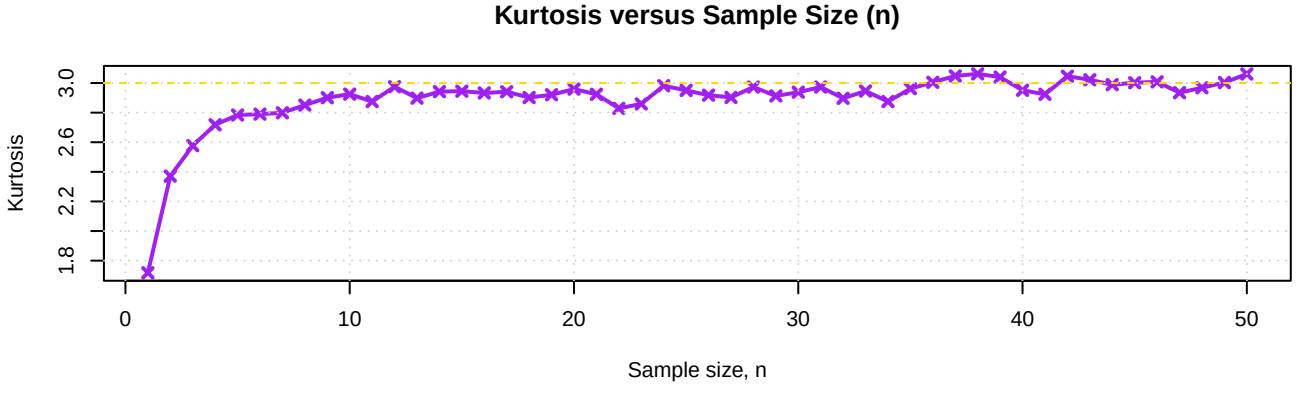

Q−Q Plot: n = 49
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 50
Density
0.0
0.5
1.0
1.5
2.5
3.0
3.5
4.0
4.5
Sample mean


Q−Q Plot: n = 50
Sample Quantiles
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Statistic value
3.0
3.2
3.4
3.6
Mean
Median
0
10
20
30
40
50
Sample size, n


Skewness versus Sample Size (n)
Skewness
−0.06
−0.02
0.02
0
10
20
30
40
50
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
1.8
2.2
2.6
3.0
0
10
20
30
40
50
Sample size, n